Bare Homogeneous Fast Fission Device Using One-Group Diffusion Theory.

by

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Finite Fast Fission Device

The Neutron Balance Equation, Its Time Dependence and the Concept of Reactivity[1]

The neutron-fission chain reaction is sustained by the action of neutrons, and when such a reaction is going on in a medium all of its characteristics can be specified by specifying the neutron population in terms of its rate of growth and its distribution in space and energy. Every neutron in such a population must eventually come to an end in one of three ways: it may be absorbed by a fissile nucleus and cause that nucleus to fission, it may be captured without fission by a fissile or non-fissile nucleus, or it may wander to one of the boundaries of the medium and leak out. Those neutrons that do cause fissions thereby cause the production of fresh neutrons; the average number produced per fission is designated by the symbol "γ." This quantity is a characteristic constant for a given species of fissile isotope at a given neutron energy. By the use of γ one can express the status of growth or decline of the neutron population in terms of the three processes cited above. There are two approaches we can take, both arriving art the same result. Under the first approach we define:

$$\frac{\text{Rate of birth of neutrons}}{\text{Rate of death of neutrons}} = \langle\gamma\rangle \cdot \frac{F}{F+C} \cdot (1-\mathscr{L}) \equiv k_{eff} \tag{1.1}$$

Here $\langle\ \rangle$ has been added to γ to indicate than an appropriate average is to be taken over the fissile species present and over the range of neutron energies. F is the fission rate in the medium and C is the neutron capture rate. The symbol $\mathscr{L}$ represents the fraction of neutrons produced which leak from the medium without causing fission or being captured.

In the Monte Carlo code, MCNP, k_{eff} is defined as:

$$k_{eff} \equiv \frac{\text{fission neutrons in generation i+1}}{\text{fission neutrons in generation i}} \tag{1.2}$$

which we write symbolically as

$$k_{eff} \equiv \frac{n_f(i+1)}{n_f(i)} \tag{1.3}$$

But since

$$n_f(i+1) = n_f(i) \cdot \langle\gamma\rangle \cdot (1-\mathscr{L}) \cdot \frac{F}{F+C} \tag{1.4}$$

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[1] From T.J. Thompson and J.G. Beckerley, *The Technology of Nuclear Reactor Safety* (MIT Press, Cambridge, 1964), Volume 1, pp.12-13 and John R. Lamarsh, *Introduction to Nuclear Engineering* (Addison-Wesley Publishing Company, Reading, Massachutts, 1975), Chapter 6.

$$k_{eff} \equiv \frac{n_f(i+1)}{n_f(i)} = \cdot \langle\gamma\rangle \cdot (1 - \mathscr{L}) \cdot \frac{F}{F + C} \qquad (1.5)$$

Thus, the definitions of k_{eff} in equations (1.1) and (1.2) above are one and the same.

In one-group diffusion theory, the term "one-group" implies that the neutrons are monoenergetic. As with regard to $\langle\gamma\rangle$ in the equation above, the model assumes that all neutrons have the same energy and therefore the same speed. This of course is not the case, which can lead to significant uncertainties in critical mass and neutron multiplication calculations. Prompt neutrons from fission are emitted with a continuous energy spectrum over several Mev, with an average energy of about 2 Mev.[2] The best fit to critical mass calculations using one-group diffusion theory is obtained by assuming the average neutron speed, v, is 1.51×10^9 cm/sec, which corresponds to an energy of 1.2 MeV.[3]

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If it is assumed that a value, τ, can be assigned which will represent adequately the average lifetime per neutron, then the dynamics of the chain-reacting system can be expressed in terms of k_{eff}. If there are n neutrons present in the system at time t, the rate of death of neutrons will be n/τ, and the rate of increase of the neutron population, dn/dt, will be:

$$\frac{dn}{dt} = \text{rate of birth of neutrons} - \text{rate of death of neutrons} \qquad (1.6)$$

$$= \frac{\text{rate of birth} - \text{rate of death}}{\text{rate of death}} \times \text{rate of death} \qquad (1.7)$$

$$= (k_{eff} - 1)\frac{n}{\tau} = \alpha n \qquad (1.8)$$

or

$$\frac{dn}{n} = \frac{(k_{eff} - 1)}{\tau} dt = \alpha\, dt \qquad (1.9)$$

which has the solution:

[2] Robert Serber, *The Los Alamos Primer*, (Berkeley: University of California Press, 1992; an annotated reprint of Serber's 1943 Los Alamos lecture notes), p. 19.

[3] Randall K. Cole, Jr. and James H. Renken, ``Analysis of the Microfission Reactor Concept," *Nuclear Science and Engineering*, 58, 345-353 (1975).

$$n = n_o \exp\{(k_{eff} - 1)(t-t_o)/\tau\} = n_o \exp[\alpha(t-t_o)] \qquad (1.10)$$

where n_o is the population at the initial time, t_o.

$(k_{eff} - 1) \equiv k_{ex}$, where k_{ex} is the symbol for "k excess." (1.11)

Since the death of each neutron, on the average, generates k_{eff} new neutrons, one may consider, instead of the average neutron lifetime, τ, the average neutron generation time, τ^*, where $\tau^* = \tau/ k_{eff}$. In this form the kinetic equations are:

$$\frac{dn}{n} = \frac{(k_{eff} - 1)}{\tau^* k_{eff}} dt = (\rho/\tau^*)\, dt \qquad (1.12)$$

$$n = n_o \exp\{(\rho/\tau^*)(t-t_o)\} \qquad (1.13)$$

If we set $t_o = 0$, the equation reduces to

$$n = n_o e^{\alpha t} = n_o e^{\rho t/\tau^*} = n_o e^{(keff - 1)t/\tau} . \qquad (1.14)$$

The quantity ρ is called "reactivity." Its meaning is:

$$\rho = \frac{(k_{eff} - 1)}{k_{eff}} \quad (1.15)$$

$$= \frac{\text{rate of birth of neutrons} - \text{rate of death of neutrons}}{\text{rate of birth of neutrons}} \quad (1.16)$$

= fraction of neutrons born which are in excess of those required to hold the population constant. (1.17)

In modeling the behavior of a reactor it is not possible to define an average lifetime or generation time which describes the neutron behavior adequately, for the reasons that the generation times of the delayed neutrons are so vastly different from those of the prompt neutrons.

For a fast fission explosive device, we are not interested in the delayed neutrons, but only in the prompt neutrons. If the delayed neutron fraction is represented by β, then the number of prompt neutrons released per fission is $\langle\gamma\rangle$ (1-β). To simplify the algebra we assume $\langle\gamma\rangle$ represents the average number of prompt neutrons, and we drop the term
(1-β).

The second approach to arrive at the same result is to again begin with the underlying principle for a nuclear chain reaction—the conservation or balance of neutrons—namely, in any given volume, the time rate of change of neutron density is equal to the rate of production minus the rate of leakage and the rate of absorption. Here, the general equation representing this balance may be written as

$$\text{Production - Leakage - Absorption} = \frac{\partial n}{\partial t}\,, \quad (1.18)$$

where $\frac{\partial n}{\partial t}$ is the time rate of change of the neutron density.
We define the neutron density as

$$n = \frac{\text{neutrons}}{\text{cm}^3}\,.$$

The product nv, expressed as neutrons per cm^2 per sec, is a significant quantity called the neutron flux and represented by

$$\varphi = nv \text{ (neutrons/cm}^2\text{-sec)}\,. \quad (1.19)$$

The rate of production of neutrons is given by $v\Sigma_f\varphi$, where

γ = the average number neutrons released per fission,
Σ_f = macroscopic fission cross section (cm^{-1}), and
$\Sigma_f\varphi$ = the number of neutrons that fission per cm^3 per sec.

For a fast fission assembly, e.g., a nuclear weapon, we are only interested in the production of prompt neutrons. If the delayed neutron fraction is represented by β, then the number of prompt neutrons is γ (1-β). To simplify the algebra, in what follows we assume v represents the average number of prompt neutrons, and we drop the term (1-β).

The rate of absorption of neutrons is given by $\Sigma_a\varphi$, where

Σ_a = macroscopic absorption cross section (cm^{-1}), and
$\Sigma_a\varphi$ = the number of neutrons that are absorbed per cm^3 per sec, and hence the absorption term will be written as - $\Sigma_a\varphi$.

In order to write the expression for the rate of leakage of neutrons in equation (1.18), we must first introduce some additional concepts of diffusion theory. The diffusion of monoenergetic neutrons may be represented by Fick's law of diffusion, according to which the net number of neutrons flowing in unit time through a unit area normal to the direction of flow, represented by the vector **J**, is given by

$$\mathbf{J} = - D_o \text{ grad } n , \qquad (1.20)$$

where **J** is called the current density, D_o is the conventional diffusion coefficient, and grad is the gradient operator. The gradient operator is often represented by the symbol ∇; consequently equation (1.20) also can be written as

$$\mathbf{J} = - D_o \nabla n . \qquad (1.21)$$

In terms of neutron flux, Fick's law is written as

$$\mathbf{J} = - D \text{ grad } \varphi , \qquad (1.22)$$

where $D = D_o/v$.

Another useful concept is that of the *mean free path*, usually denoted by λ. Although this symbol is also used for the disintegration constant, there will be no confusion because the mean free path λ will always have a subscript denoting a particular process. The mean free path for a process is related to the cross section for that process; thus the average distance a neutron moves between scattering collisions is called the *scattering mean free path* λ_s, defined by the relation

$$\lambda_s(\text{cm}) \equiv 1/\Sigma_s = 1/(N\sigma_s) , \qquad (1.23)$$

where N is nuclei density (the number of nuclei per cm^3), and Σ_s and σ_s are the *macroscopic scattering cross section* and the *microscopic scattering cross section*, respectively. The latter is the scattering cross section per nuclei. There is also a *transport mean free path* defined as

$$\lambda_{tr} \equiv 1/\Sigma_{tr} \equiv \frac{1}{N\sigma_{tr}} \equiv \frac{1}{\Sigma_s(1-<\mu_o>)} , \tag{1.24}$$

where $<\mu_o> = <\cos\theta>$ is the average value of the scattering angle in the laboratory reference system. It can be shown that

$$<\mu_o> \equiv <\cos\theta> = \frac{2}{3A} , \tag{1.25}$$

where A is the mass number of the scattering nucleus. For uranium and plutonium, with mass numbers in the range 235-240:

	$<\mu_o>$	$(1-<\mu_o>)$
U,Pu	0.0028	0.9972

The transport mean free path is introduced in order to take into account the fact that the scattering is preferentially in the forward direction; it is greater than the scattering mean free path, which means that, on average, a neutron will travel farther in a given number of collisions than if there were no preferred direction. The microscopic transport cross section, $\sigma_{tr} = \sigma_s(1-<\mu_o>)$, is then a measure of the rate at which the neutron loses its forward momentum or the ``memory'' of its original direction. According to kinetic theory, the diffusion coefficient is given by

$$D = \frac{\lambda_{tr}}{3} = \frac{1}{3\Sigma_{tr}} . \tag{1.26}$$

Hence, if the neutron velocity is known, λ_{tr} is determined from equation (1.24) in terms of the properties N, $<\mu_o>$, and σ_s of the medium in which the neutrons are diffusing, and then the rate of diffusion can be calculated from equation (1.26).

But equation (1.26) is holds only for weakly-absorbing media, that is, when

$\Sigma_a << \Sigma_s$. For moderate absorbers it can be shown from transport theory that the diffusion approximation is still satisfactory in regions which are more than two or three mean free paths away from boundaries and neutron sources, provided that the diffusion coefficient D, correct to first order terms in Σ_a/Σ_t, is represented by

$$D = \frac{1}{3\Sigma_t(1-\langle\mu_o\rangle)(1-(4/5)(\Sigma_a/\Sigma_t)+(\Sigma_a/\Sigma_t)(\langle\mu_o\rangle/(1-\langle\mu_o\rangle))+...)}, \qquad (1.27)$$

where Σ_t and Σ_a are the total and absorption macroscopic cross sections, respectively. The diffusion approximation holds in regions away from boundaries and sources, because the angular distribution of neutron velocity vectors does not then depend strongly on position. Unfortunately, this is not the case for small nuclear weapon assemblies. The bare critical radii of the Godiva (HEU) and Jezebel (WGPu) critical assemblies was estimated to be 0.56 λ_{is} and 0.17 λ_{is}, respectively, where λ_{is} is the inelastic scattering mean free path.[4]

Returning to the equation for the conservation of neutrons, the net rate of outflow of neutrons per unit volume is equal to the divergence of the vector **J**, i.e., div **J**, and hence

$$\begin{aligned}\text{Neutron leakage per unit volume per second} &= -\,\mathrm{div}\, D\nabla\varphi \\ &= -\,D\nabla^2\varphi,\end{aligned} \qquad (1.28)$$

where ∇^2 is the Laplacian operator. The neutron balance equation (1.18), thus, can be variously written as

$$D\nabla^2\varphi + \gamma\Sigma_f\varphi - \Sigma_a\varphi = \frac{\partial n}{\partial t}, \qquad (1.29)$$

or

$$D\nabla^2\varphi + [\gamma\Sigma_f - \Sigma_a]\varphi = \frac{1}{v}\frac{\partial\varphi}{\partial t}, \qquad (1.30)$$

or

$$D\nabla^2 n + [\gamma\Sigma_f - \Sigma_a]n = \frac{1}{v}\frac{\partial n}{\partial t}. \qquad (1.31)$$

The macroscopic absorption cross section, Σ_a, is the sum of the macroscopic fission cross section, Σ_f, plus the macroscopic non-fission capture cross section, Σ_c. Thus, the term $[\gamma\Sigma_f - \Sigma_a]$ can also be written as

$$[\gamma\Sigma_f - \Sigma_a] = [(\gamma-1)\Sigma_f - \Sigma_c]. \qquad (1.32)$$

[4] Leona Stewart, "Leakage Neutron Spectrum for a Bare Pu^{239} Critical Assembly," *Nuc. Sci. Eng.*, 8, 595-597 (1960).

To solve the differential equation for the conservation of neutrons, φ is assumed to have the form

$$\varphi(\mathbf{r},t) = \Phi_{\mathbf{r}}(\mathbf{r})\Phi_t(t) \tag{1.33}$$

and

$$\Phi_t(t) = e^{\alpha t} . \tag{1.34}$$

Thus,

$$D\nabla^2\Phi_{\mathbf{r}} + [\gamma\Sigma_f - \Sigma_a - (\alpha/v)]\Phi_{\mathbf{r}} = 0 , \tag{1.35}$$

or

$$D\nabla^2\varphi + [\gamma\Sigma_f - \Sigma_a - (\alpha/v)]\varphi = 0 . \tag{1.36}$$

Buckling. Letting

$$DB^2 = [\gamma\Sigma_f - \Sigma_a - (\alpha/v)] , \tag{2.1}$$

where the parameter B is called the "buckling," equation (1.36) reduces to the wave equation:

$$\nabla^2\varphi + B^2\varphi = 0 . \tag{2.2}$$

The neutron balance equation (1.30) can be rewritten

$$D(-B^2)\varphi + [\gamma\Sigma_f - \Sigma_a]\varphi = \frac{1}{v}\frac{\partial\varphi}{\partial t} , \tag{2.3}$$

or

$$\frac{1}{v}\frac{\partial\varphi}{\partial t} = [\gamma\Sigma_f - \Sigma_a - DB^2]\varphi , \tag{2.4}$$

or

$$\frac{1}{v\gamma\Sigma_f}\frac{\partial\varphi}{\varphi} = [1 - \frac{\Sigma_a + DB^2}{\gamma\Sigma_f}]\,\partial t , \tag{2.5}$$

or

$$\tau^* \frac{\partial\varphi}{\varphi} = \frac{k-1}{k}\,\partial t , \tag{2.6}$$

where

$$\tau^* \equiv \frac{1}{v\gamma\Sigma_f} , \tag{2.7}$$

and

$$k_{eff} = k \equiv \frac{\gamma\Sigma_f}{\Sigma_a + DB^2} . \tag{2.8}$$

Because k is the multiplication of a particular system and k = 1 always implies criticality, the difference $k_{ex} \equiv k - 1$ is the "excess multiplication" with respect to a critical condition. The symbol for excess multiplication, k_{ex}, is sometimes written Δk. The term (k - 1)/k then represents the "fractional excess multiplication." This defines the reactivity:[5]

$$\rho \equiv \frac{k-1}{k} = \frac{k_{ex}}{k}, \tag{2.9}$$

Equation (2.6) is also written as

$$\frac{\partial\varphi}{\varphi} = \frac{\rho}{\tau^*}\,\partial t, \tag{2.10}$$

which has the solution

$$\varphi(\mathbf{r},t) = \Phi_{\mathbf{r}}(\mathbf{r})\Phi_t(t) = \Phi_{\mathbf{r}}(\mathbf{r})\, e^{\rho t/\tau^*}. \tag{2.11}$$

The exponential factor in equation (2.11) is alternative means of expressing α commonly used in reactor theory:

$$\alpha = \rho/\tau^* = k_{ex}/\tau$$

where $\tau = k\tau^*$. (2.12)

Infinite Homogeneous Medium. In an infinite homogeneous medium, during any given period just as many neutrons leak into a given volume as leak out. In other words, the net rate of neutron leakage per unit volume is zero, $- D\nabla^2\varphi = DB^2 = 0$. Therefore,

$$\alpha_\infty = v[\gamma\Sigma_f - \Sigma_a] = vN[\gamma\sigma_f - \sigma_a]. \tag{3.1}$$

Also, for an infinite medium equation (2.8) reduces to

$$k_\infty \equiv \gamma(\Sigma_f/\Sigma_a) = \gamma(\sigma_f/\sigma_a). \tag{3.2}$$

Therefore, we can also express α_∞ as

$$\alpha_\infty = \frac{[\gamma(\sigma_f/\sigma_a) - 1]}{\tau_a} = \frac{[k_\infty - 1]}{\tau_\infty}, \tag{3.3}$$

where

$$\tau_a \equiv \tau_\infty \equiv \frac{1}{v\Sigma_a}. \tag{3.4}$$

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[5] Not to be confused with density which utilizes the same Greek symbol.

If we define $\tau_{a,o} \equiv \tau_a(\rho_o)$, i.e., τ_a at normal density, then $\tau_a = \tau_{a,o}/\eta$, where $\eta \equiv \rho/\rho_o$; and

$$\alpha_\infty = \frac{[\gamma(\sigma_f/\sigma_a) - 1]}{\tau_{a,o}}\,\eta = \alpha_{\infty,o}\,\eta\ . \tag{3.5}$$

General Expressions for the Neutron Multiplication Factor, α. From equation (2.1), α can also be written:

$$\alpha = v\,[\gamma\Sigma_f - \Sigma_a - DB^2]\ , \tag{4.1}$$

$$= [\gamma(\sigma_f/\sigma_a) - 1]/\tau_a - D_oB^2\ , \tag{4.2}$$

$$= [k_\infty - 1]/\tau_a - D_oB^2\ , \tag{4.3}$$

$$= \alpha_\infty - D_oB^2\ . \tag{4.4}$$

Spherically Symmetric Bare Homogeneous Device. In spherical coordinates $\Phi_{\mathbf{r}}(\mathbf{r})$ in equations (1.33), (1.35) and (2.11), takes the form $\Phi_{\mathbf{r}}(\mathbf{r}) = \Phi_r(r)\Phi_\theta(\theta)\Phi_\varphi(\varphi)$. In the case of spherical symmetry the two angular components, $\Phi_\theta(\theta)$ and $\Phi_\varphi(\varphi)$, equal unity, and equation (1.35) reduces to the radial component

$$D\nabla^2\Phi_r + [v\Sigma_f - \Sigma_a - (\alpha/v)]\Phi_r = 0\ , \tag{5.1}$$

where we have replaced the vector **r**, by the scalar r, and the Laplacian operator, ∇^2, in spherical coordinates now includes only its radial component:

$$\nabla^2 = \frac{\partial}{\partial r^2} + \frac{2}{r}\frac{\partial}{\partial r}\ . \tag{5.2}$$

The solution to equation (5.1) takes the form

$$\Phi_r = A\,\frac{\sin(\pi r/R)}{r}\ . \tag{5.3}$$

Combining equations (5.1), (5.2) and (5.3), and rearranging terms, we have

$$[\gamma\Sigma_f - \Sigma_a - (\alpha/v) - (D\pi^2/R^2)]\,A\,\frac{\sin(\pi r/R)}{r} = 0\ . \tag{5.4}$$

The left-hand side of equation (5.4) must equal zero, which implies

$$\alpha = v\,[\gamma\Sigma_f - \Sigma_a - D(\pi/R)^2)]\ . \tag{5.5}$$

Comparing equations (2.1) and (5.5), we see that the factor (π/R) is the buckling for the special case of spherical symmetry.

Since $D = D_o/v$, and $\tau_a = 1/v\Sigma_a$, equation (5.5) can be written

$$\alpha = [\gamma(\sigma_f/\sigma_a)-1]/\tau_a - D_o(\pi/R)^2 , \tag{5.6}$$

$$= \alpha_\infty - D_o(\pi/R)^2 . \tag{5.7}$$

The critical radius, R_c, is defined as the radius at which $\alpha = 0$; therefore from equation (5.7)

$$R_c^2 = \frac{\pi^2 D_o}{\alpha_\infty} . \tag{5.8}$$

$$R_c^2 = \frac{\pi^2 D_o \tau_a}{[\gamma(\sigma_f/\sigma_a)-1]} = \frac{\pi^2 D}{N[\gamma\sigma_f - \sigma_a]} . \tag{5.9}$$

Substituting D, from equation (1.27), into equation (5.9), and rearranging terms gives

$$(R_c\rho)^2 = \frac{\pi^2}{3(N_A/A)\sigma_t[\gamma\sigma_f - \sigma_a](1-\langle\mu_o\rangle)(1-(4/5)(\sigma_a/\sigma_t)+(\sigma_a/\sigma_t)(\langle\mu_o\rangle/(1-\langle\mu_o\rangle))+...)} , \tag{5.10}$$

where we have made use of the fact that $N = \rho N_A/A$, where N_A ($= 0.602217\times10^{24}$ nuclei/mol) is *Avogadro's Number*, and A (g/mol) is the atomic mass number. The right-hand side of equation (5.10) is a constant for a given material, therefore

$$(\rho R)_c = \text{a constant.} \tag{5.11}$$

Since the *critical mass*, $M_c = (4/3)\pi\rho R_c^3$, from equation (5.10)

$$M_c = (A/3N_A)^{3/2} (\pi^4/3) (1/\rho)^2 F^{3/2} , \tag{5.12}$$

where

$$F = \sigma_t[\gamma\sigma_f - \sigma_a](1-\langle\mu_o\rangle)(1-(4/5)(\sigma_a/\sigma_t)+(\sigma_a/\sigma_t)(\langle\mu_o\rangle/(1-\langle\mu_o\rangle))+...) . \tag{5.13}$$

From equation (5.12), we see that for a given material

$$M_c \propto (1/\rho)^2 . \tag{5.14}$$

For plutonium and uranium, $F \approx (\sigma_t - 0.8\sigma_a)[\gamma\sigma_f - \sigma_a] = (\sigma_s + 0.2\sigma_a)[\gamma\sigma_f - \sigma_a]$.

Substituting equation (5.8) into (5.7), gives

$$\alpha = \alpha_\infty [1- (R_c /R)^2] \qquad = \alpha_{\infty,o}\, \eta\, [1- (R_c /R)^2] , \tag{5.15}$$

$$\alpha = \frac{(k-1)}{\tau_a} [1- (R_c /R)^2] \qquad = \frac{(k-1)}{\tau_{a,o}}\, \eta\, [1- (R_c /R)^2] , \tag{5.16}$$

$$\alpha = \frac{[\gamma(\sigma_f /\sigma_a) - 1]}{\tau_a} [1- (R_c /R)^2] = \frac{[\gamma(\sigma_f /\sigma_a) - 1]}{\tau_{a,o}}\, \eta\, [1- (R_c /R)^2] . \tag{5.17}$$

If we define the number of "crits," $\kappa \equiv M/M_c$, the ratio of the mass to the critical mass, then

$$\kappa = (R/R_c)^3 = \kappa_o\, \eta^2 , \text{ where } \kappa_o = M/M_{c,o} \tag{5.18}$$

and

$$\alpha = \alpha_\infty [1 - \kappa^{-2/3}] = \alpha_{\infty,o}\, \eta\, [1 - \kappa^{-2/3}] . \tag{5.19}$$

Also, if we let

$x = \rho R$
$y = \alpha R$
$a = \alpha_{\infty,o} /\rho_o$
$b = (\rho R)_c$

then

$$y = ax[1-(b/x)^2] \tag{5.20}$$

Summary of terms:

n	neutron density (neutrons/cm^3)	(s.1)
v	average neutron speed (cm/sec)	(s.2)
N	nuclei density (nuclei/cm^3)	(s.3)
ρ	density (g/cm^3)	(s.4)
ρ_o	normal density (g/cm^3)	(s.5)
$\rho \equiv (k-1)/k$	reactivity	(s.6)
$\eta \equiv \rho/\rho_o$	ratio of the density to the normal density	(s.7)
$\varphi = nv$	neutron flux (neutrons/cm^2-sec)	(s.8)
$\varphi(r,t) = \Phi_r(r)\Phi_t(t)$	neutron flux (neutrons/cm^2-sec)	(s.9)
$\Phi_r = \dfrac{\sin(\pi r/R)}{r}$	radial component of the flux distribution for a bare homogeneous sphere	(s.10)
$\Phi_t(t) = e^{\alpha t}$	time dependence of the flux distribution	(s.11)
D_o	conventional diffusion coefficient (cm^2/sec)	(s.12)
$D = D_o/v$	diffusion coefficient (cm)	(s.13)
$B = \pi/(R + \delta)$	Buckling for a sphere, where δ is the extrapolation distance.	(s.14)
$L = (D/\Sigma_a)^{1/2}$	Diffusion length (cm)	(s.15)
$L^2B^2 = \dfrac{DB^2}{\Sigma_a} = D_o\tau_a B^2$	$\dfrac{\text{leakage}}{\text{absorption}}$	(s.16)
$k_\infty = \gamma(\sigma_f/\sigma_a) = \tau_a/\tau^*$	neutron multiplication in an infinite medium	(s.17)
$k = \dfrac{k_\infty}{(1+L^2B^2)}$	neutron multiplication in an finite medium	(s.18)
$k = \dfrac{\gamma\Sigma_f}{(\Sigma_a+DB^2)}$	neutron multiplication in an finite medium	(s.19)

$k_{ex} \equiv k - 1$	excess multiplication	(s.20)
$\alpha_{\infty,o} = \dfrac{[k_\infty - 1]}{\tau_{a,o}}$	alpha for an infinite medium at normal density (sec^{-1}, or shakes^{-1})	(s.21)
$\alpha_{\infty,o} = \dfrac{[\gamma(\sigma_f/\sigma_a) - 1]}{\tau_{a,o}}$	alpha for an infinite medium at normal density (sec^{-1}, or shakes^{-1})	(s.22)
$\alpha_\infty = \alpha_{\infty,o}\, \eta$	alpha for an infinite medium at density ρ	(s.23)
$\alpha_\infty = vN[\gamma\sigma_f - \sigma_a]$	alpha for an infinite medium at density ρ	(s.24)
$\alpha_\infty = vN[(\gamma -1)\sigma_f - \sigma_c]$	alpha for an infinite medium at density ρ	(s.25)
$\alpha = \rho/\tau^* = k_{ex}/\tau = (k-1)/\tau$	alpha for a finite medium (sec^{-1}, or shakes^{-1})	(s.26)
$\alpha = \alpha_{\infty,o}\, \eta\, [1- (R_c/R)^2]$	alpha for a finite medium (sec^{-1}, or shakes^{-1})	(s.27)
$\alpha = \alpha_\infty\, [1- (R_c/R)^2]$	alpha for a finite medium (sec^{-1}, or shakes^{-1})	(s.28)
$\tau_a = \tau_\infty = 1/v\Sigma_a$	mean (average) lifetime of a neutron in an infinite medium (sec, or shakes)	(s.29)
$\tau_{a,o}$	τ_a for material at normal density	(s.30)
$\tau_a = \tau_{a,o}/\eta$	mean (average) lifetime of a neutron in an infinite medium (sec, or shakes)	(s.31)
$\tau_f = 1/v\Sigma_f$	mean (average) lifetime of a neutron for fission	(s.32)
$\tau^* = 1/v\gamma\Sigma_f = \gamma\tau_f = \tau_a/k_\infty$	mean (average) time between fission neutron productions.	(s.33)
$\tau = \dfrac{\tau_a}{(1 + L^2B^2)}$	mean (average) lifetime of a neutron in a finite medium	(s.34)
$\tau = k\, \tau^*$	mean (average) lifetime of a neutron in a finite medium. At prompt critical, k=1, and $\tau = \tau^*$.	(s.35)
τ_o	mean (average) lifetime of a neutron in a finite medium at normal density	(s.36)

Comparing Serber's Nomenclature:

Term	**Serber**	**This Report**
Current Density	D	D_o
Transport mean free path	l	λ_{tr}
Transport microscopic cross section	σ_t	σ_{tr}
Neutron Density (neutrons/ cm^2)	N (p. 25)	n
Nucleii density (nuclei/cm^2)	n	N
Alpha	γ'/τ	α
Excess multiplication factor	γ'	$k_{ex} = (k_{eff} - 1)$
Mean lifetime of a neutron in an infinite (this report), or a finite (Serber?) medium[6]	τ	τ_a

Under this report's derivation:
$\tau = \tau_a/(1+ L^2B^2)$, where τ_a is the mean lifetime of a neutron in an infinite medium

$\tau_a = 1/v\Sigma_a$

Number of neutrons produced per neutron absorbed	γ	$\gamma\sigma_f/\sigma_a = k_\infty$
Number of neutrons produced per fission	$\gamma\sigma_a/\sigma_f$	γ

Serber's (p. 26) γ'/τ is the same as α, and since

$\gamma'/\tau = \alpha = \rho/\tau^* = (k_{eff} - 1)/\tau = k_{ex}/\tau. = k_{ex}/\tau$

Then Serber's γ' is the same as $(k_{eff} - 1) = k_{ex}$, but his τ is the same as τ_a, the average lifetime per neutron in an infinite medium.

[6] This can be seen be comparing Serber's (p. 25) net rate of generation of neutrons per unit volume, $(\gamma - 1)\, N/\tau$ with Cochran's rate of production of neutrons per unit volume in an infinite medium, $\gamma\, \Sigma_a\varphi$. Thus, $\gamma_s/\tau = \gamma_c\, \Sigma_f\, v$, or $\tau = (\gamma_c/\gamma_s)(1/v\Sigma_f) = (\sigma_a/\sigma_f)\tau_f = \tau_a$.

Calculation of the Efficiency of a Pure Fission Weapon.

The following derivation of the efficiency of a nuclear fission device parallels that of Serber,[7] but makes fewer approximations.

The derivation assumes that all of the fission energy released during the chain reaction is converted into internal energy, and the internal energy, equivalent to that produced during one neutron generation, is sufficient to drive the system subcritical. This is equivalent to saying the expanding core, from its state of maximum supercriticality to criticality ($\alpha = 0$), takes place during one neutron generation ($\Delta t = 1/\alpha$). The model assumes that approximately one-half of the energy released by the fission chain reaction occurs when $\alpha \geq 0$, and the remaining one-half is released as α falls from 0 to -1. The model assumes that the fission energy is added instantaneously to the system before spherically-symmetric homogeneous isentropic expansion begins.

Homogeneous expansion implies that the density, although time dependent, is independent of the spacial variables. It will be shown that at any given time the velocity of each mass shell is proportional to the radius. The average pressure is assumed to be time independent during the expansion phase.

In the derivation the total mass remains constant, but we need to distinguish the normal (uncompressed), initial (compressed), and final (expanded) densities and radii, and the critical masses and radii of each state. Our convention will be to let the subscripts o, i, and c refer to the uncompressed, initial (fully compressed) and final (critical) states, respectively; and M and R will refer to the actual mass and radius, and the italics *M* and *R* will refer to the respective critical mass and radius. In sum, we use the following variables:

M : core mass (a constant)
M_o : the critical mass at normal density
M_i : the critical mass at $R = R_i$, when fully compressed
M_c : the critical mass at $R = R_c$

r(s,t) : radius of mass-shell labeled s at time t
u(s,t) : velocity
p(s,t) : pressure
<p(t)> : average pressure
ρ(t) : density
R(t) : outer radius of spherical mass M at time t
U(t) : velocity of the outer radius

[7] Serber, *The Los Alamos Primer*, (1992), pp. 38-43 (see particularly p. 42).

$\kappa_o \equiv M/M_o = (R_o/R_o)^3$: the number of critical masses at normal density
$\eta \equiv \rho_i/\rho_o$: the degree of compression
$\kappa \equiv M/M_i$: the number of critical masses when fully compressed,
i.e., the number of "crits".

E_k : kinetic energy
E_i : internal energy
Y : fission yield
E_{ff} : efficiency

We begin by examining the velocity and pressure profiles of the internal mass-points, or mass-shells, as a function of radius.

Velocity profile. Assume $\rho = \rho(t)$, and not a function of r

$$M = \frac{4}{3} \pi R^3 \rho = \text{constant} . \qquad (6.1)$$

We assume the mass is comprised of the sum of individual thin shells, like an onion, and examine two mass-shells having initial radii r_1 and r_2.

$$dm_1 = 4\pi\, r_1^2 \rho_o\, dr, \text{ and } dm_2 = 4\pi\, r_2^2 \rho_o\, dr , \qquad (6.2)$$

therefore

$$\frac{dm_2}{dm_1} = \frac{r_2^2}{r_1^2} . \qquad (6.3)$$

Assume dm_1 has velocity v_1 and in time Δt moves to $r_1 + v_1\Delta t$, and dm_2 has velocity v_2 and in time Δt moves to $r_2 + v_2\Delta t$. Therefore,

$$\frac{dm_2}{dm_1} = \frac{(r_2^2 + v_2\Delta t)^2}{(r_1^2 + v_1\Delta t)^2} = \frac{r_2^2}{r_1^2} , \qquad (6.4)$$

or

$$r_1^2 r_2^2 + 2r_1^2 r_2 v_2 \Delta t + r_1^2 v_{22} \Delta t^2 = r_1^2 r_2^2 + 2r_1 r_2^2 v_1 \Delta t + r_{22} v_1^2 \Delta t^2 . \qquad (6.5)$$

In the limit as $\Delta t \to 0$, the Δt^2 terms are dropped and the equation reduces to

$$r_1 v_2 = r_2 v_1 , \quad \text{or} \quad \frac{v_2}{v_1} = \frac{r_2}{r_1} . \qquad (6.6)$$

Thus, at any time t the velocity of a mass-shell is proportional to the radius of the shell. It is convenient to define the position of a mass-point (or mass-shell) using a new coordinate, s, where

$$r(s,t) = s\,R(t)\ , \tag{6.7}$$

or

$$s \equiv r(t)/R(t), \text{ and } r(1,t) = R(t)\ . \tag{6.8}$$

The velocity of the mass-shell is given by

$$u(s,t) = r\,(s,t) = sR\,(t) = s\,U(t)\ . \tag{6.9}$$

Pressure profile. To calculate the pressure profile we turn to Newton's second law, F = ma, which for a mass-shell can be written

$$\rho\, 4\pi\, r^2\, drr\ = 4\pi\, r^2 \left(-\frac{dp}{dr}\right) dr\ , \tag{6.10}$$

or

$$\rho r\ = -\frac{dp}{dr}\ . \tag{6.11}$$

Since

$$r = s\,Rr \qquad \dot{r} = s\dot{R}\,r \qquad \ddot{r} = s\ddot{R} \tag{6.12}$$

$$\frac{dp}{dr} = \frac{dp}{ds}\frac{ds}{dr} = \frac{1}{R}\frac{dp}{ds} \tag{6.13}$$

then,

$$\rho u = \rho\ddot{r} = \rho s\ddot{R} = -\frac{dp}{dr} = -\frac{1}{R}\frac{dp}{ds}\ , \tag{6.14}$$

or

$$\frac{dp}{ds} = -\,\rho sR\ddot{R}\ . \tag{6.15}$$

Assume p takes the form

$$p(s,t) = -\frac{\rho s^2 R\ddot{R}}{2} + f(t)\ . \tag{6.16}$$

The boundary conditions are at s = 0, p(s,t) = p(0,t) = f(t), which gives

$$p(s,t) = p(0,t) - \frac{\rho s^2 R\ddot{R}}{2}\ , \tag{6.17}$$

and at s = 1, r = R and p(1,t) = 0. Therefore,

$$p_o(t) \equiv p(0,t) = \frac{\rho R\ddot{R}}{2} , \qquad (6.18)$$

or

$$\rho\ddot{R} = 2p_o/R , \qquad (6.19)$$

and

$$p(s,t) = p_o(t)\,(1-s^2) , \qquad (6.20)$$

or

$$p(r,t) = p_o(t)\,[1-(r/R)^2] . \qquad (6.21)$$

The average pressure at time t, <p(t)>, is given by

$$\langle p(t)\rangle = p_o(t)\,\frac{\int p(r,t)\,dV}{\int dV} . \qquad (6.22)$$

With p(r,t) from equation 3.21, this becomes

$$\langle p(t)\rangle = p_o(t)\,\frac{\int_o^R r^2dr - \int_o^R (r^4/R^2)dr}{\int_o^R r^2dr} = \frac{(R^3/3) - (R^3/5)}{(R^3/3)} , \qquad (6.23)$$

$$\langle p(t)\rangle = (2/5)\,p_o(t). \qquad (6.24)$$

Kinetic energy. The kinetic energy, E_k, of the expanding core is given by

$$E_k = \int_o^R \tfrac{1}{2}\dot{r}^2\,\rho\,4\pi r^2dr = 2\pi\rho R^3\dot{R}^2 \int_o^1 s^4\,ds , \qquad (6.25)$$

$$= \tfrac{1}{2}\,(3M/5)\dot{R}^2 . \qquad (6.26)$$

Thus, the spherical homogenous expansion can be treated as a thin spherical shell of mass, (3/5)M and radius R. This also is seen from Newton's Second Law, which from equation 3.18 can be written:

$$M\ddot{R} = 2Mp_o/(\rho R) = 2Vp_o/R = (8\pi/3)R^2p_o , \qquad (6.27)$$

or

$$(3/5)M\ddot{R} = 4\pi R^2\langle p(t)\rangle , \qquad (6.28)$$

where $4\pi R^2$ is the area of the mass-shell and <p(t)> the average pressure.

The time rate of change of the kinetic energy is

$$\frac{dE_k}{dt} = (3M/5)\dot{R}\ddot{R} = (3/5)\, M\dot{R}\ (2p_o/\rho R)\ , \tag{6.29}$$

$$= (2/5)\ p_o\ 4\pi R^2\dot{R}\ . \tag{6.30}$$

Since $V = (4/3)\pi R^3$ and $\dot{V} = 4\pi R^2\dot{R}$, then

$$\dot{E}_k = (2/5)\ p_o\dot{V}\ , \tag{6.31}$$

$$= \langle p(t)\rangle\dot{V}\ . \tag{6.32}$$

Equation (6.28) is equal to the integral of equation (6.32) with respect to time:

$$E_k = (3/10)\ M\dot{R}^2 = \int_{V_i}^{V_f} \langle p(t)\rangle\ dV. \tag{6.33}$$

We assume the average pressure is slowly varying with time, therefore

$$E_k = (3/10)\ M\dot{R}^{\,2} \approx \langle p(t)\rangle \int_{V_i}^{V_f} dV = \langle p(t)\rangle\ \Delta V\ . \tag{6.34}$$

Internal energy and yield. The internal energy, E_i, is given by

$$E_i = \frac{\langle p\rangle\ V}{(\gamma\text{-}1)}\ , \tag{6.35}$$

where $\gamma = 5/3$ for a monatomic Boltzmann gas, or a Fermi-degenerate plasma. Since we assume that about one-half of the fission energy released occurs after α fall below zero, therefore

$$Y \approx 2\ E_i = \frac{\langle p\rangle\ V}{(\gamma\text{-}1)} \approx \frac{2\ E_k}{(\gamma\text{-}1)}\ \frac{V}{\Delta V} = (9/10)\ M\dot{R}^2\ \frac{V}{\Delta V}\ . \tag{6.36}$$

This is (9/10) the value used by Serber.[8] We approximate

$$\dot{R} = \Delta R/\Delta t \approx (R_c\text{-}R_i)\ \ \alpha\ , \tag{6.37}$$

where $R_c = \mathcal{R}_c$ is the final (i.e., critical) radius, and R_i is the initial (i.e., maximum compressed) radius, respectively.

[8] Serber, *The Los Alamos Primer*, (1992), p. 42.

Since the critical mass $M \propto (1/\rho)^2$,

$$\kappa = M/M_i = (M/M_o)(M_o/M_i) = \kappa_o(\rho_i/\rho_o)^2 , \tag{6.38}$$

or

$$\kappa = \kappa_o\,\eta^2 = M/M_i = (R/R_i)^3 . \tag{6.39}$$

Also, since $R \propto (1/\rho)$ and $R_c = R_c$,

$$R_c/R_o = \rho_o/\rho_c ;\;\; R_c = R_o(R_c/R_o)^3 ;\;\; R_c/R_o = (R_o/R_o)^{1/2} = \kappa_o^{1/6} \tag{6.40}$$

Now we return to $\Delta R \approx (R_c\text{-}R_i)$, and rewrite it as

$$\Delta R \approx R_o[(R_c/R_o)-(R_i/R_o)] , \tag{6.41}$$

$$= R_o\,\kappa_o^{1/3}(\kappa_o^{1/6} - \eta^{-1/3}) , \tag{6.42}$$

$$= (R_o/\eta)\,\kappa^{1/3}(\kappa^{1/6} - 1) . \tag{6.43}$$

Also, from equation (5.19)

$$\alpha = \alpha_\infty(1\text{-}\kappa^{-2/3}) , \tag{6.44}$$

where $\alpha_\infty = \alpha_{\infty,o}\,\eta$, and $\alpha_{\infty,o} = [v(\sigma_f/\sigma_a)\text{-}\ 1]/\tau_{a,o}$. (6.45)

Thus, $\dot{R} = \Delta R/\Delta t \approx (R_c\text{-}R_i)\,\alpha$ is given by

$$\dot{R} = R_o\;\alpha_{\infty,o}\,\kappa^{1/3}(\kappa^{1/6} - 1)(1\text{-}\kappa^{-2/3}) \tag{6.46}$$

To calculate the yield, we also need V/dV, which can be written as

$$V/dV = R_c^3/(R_c^3\text{-}R_i^3) = [1\text{-}(R_i/R_c)^3]^{-1} . \tag{6.47}$$

Since $R_i/R_c = (R_i/R_o)(R_o/R_c) = \eta^{-1/3}\kappa_o^{-1/6} = \kappa^{-1/6}$,

$$V/dV = (1\text{-}\kappa^{-1/2})^{-1} . \tag{6.48}$$

Therefore, substituting equations (6.45) and (6.48) into equation (6.36) gives:

$$Y = (9/10)\,M\,(R_o\alpha_{\infty,o})^2\,\kappa^{2/3}(\kappa^{1/6} - 1)^2(1\text{-}\kappa^{-2/3})^2(1\text{-}\kappa^{-1/2})^{-1} \tag{6.49}$$

$$= (9/10)\,M\,(R_o\alpha_{\infty,o})^2\,f(\kappa) , \tag{6.50}$$

where

$$f(\kappa) = \frac{(\kappa^{2/3} - 1)^2(\kappa^{1/6} - 1)^2}{\kappa^{1/6}(\kappa^{1/2} - 1)} \tag{6.51}$$

The function f(κ) can be written as the product

$$f(\kappa) = g(\kappa)\, h(\kappa) \;\;, \tag{6.52}$$

where

$$g(\kappa) = (\kappa^{1/3} - 1)^3 \;\;, \tag{6.53}$$

and letting $x = \kappa^{1/6}$,

$$h(x) = \frac{(x^4-1)^2(x-1)^2}{x(x^3-1)(x^2-1)^3} \;\;. \tag{6.54}$$

But g(κ) has been chosen such that h(κ) does not vary significantly with κ. In fact h(κ) is a very slowly decreases function of κ with values

κ	h(κ)
1	0.6667
2	0.6338
20	0.5899

The variability in h(κ) can be reduced even further by letting g(κ) in equation (6.52) equal

$$g(\kappa) = (\kappa^{0.302} - 1)^3 \;\;, \tag{6.55}$$

in which case

$$h(\kappa) = 0.88 \pm 0.02 \;\; \text{for} \;\; 1 \le \kappa \le 20 \;\;. \tag{6.56}$$

However, the accuracy of the model does not warrant this degree of fine tuning. Therefore, in sum,

$$Y \approx 0.55\, M\, (R_o\alpha_{\infty,o})^2\, (\kappa^{1/3} - 1)^3 \;, \tag{6.57}$$

If M, R, and α are measures in kilograms, centimeters, and shakes^{-1}, respectively, then the right hand side of the equation (6.57) must contain an additional conversion factor equal to 10^{12}, if Y is expressed in joules, or 0.24 if Y is expressed in kilotons.[9]

M_o, R_o, and $\alpha_{\infty,o}$ for bare weapon-grade plutonium and uranium are given in Table 3.1. Substituting these values into equation 6.57, the yields of bare weapon-grade plutonium and uranium assemblies predicted by the model are:

$$Y \approx a\, \kappa_o\, (\kappa^{1/3} - 1)^3 \text{ kt}, \tag{6.58}$$

where a = 300 for bare αWGPu, 480 for δWGPu, and 630 for bare WGU.

Efficiency. The efficiency, E_{ff} is the yield divided by the yield if 100 percent of the core fissioned. Therefore, from equation (6.57):

$$E_{ff} \approx (0.55/\varepsilon)\, (R_o\alpha_{\infty,o})^2\, (\kappa^{1/3} - 1)^3, \tag{6.59}$$

where ε is the energy conversion factor. If R, and α are measures in centimeters, and sec^{-1}, respectively, then $\varepsilon = 7.3\text{x}10^{17}$ ergs/g; and if α is measured in shakes^{-1}, $\varepsilon = 73$ $(\text{sec/shakes})^2$(ergs/g).[10] Substituting the values from Table 1 into equation 3.59, the efficiency of bare weapon-grade plutonium and uranium assemblies predicted by the model is:

$$E_{ff} \approx b\, (\kappa^{1/3} - 1)^3, \tag{6.60}$$

where b = 1.4 for bare WGPu, 1.2 for bare U-233 and 0.7 for bare WGU.

Except for small differences in the constants in front, equations (6.59) and (6.60) are the same as those derived by Serber for the special case of uncompressed HEU, where Serber's $\Delta = (\kappa^{1/3} - 1)$ above.[11] Serber claims that a more accurate calculation only alters the value of b in equation (6.60), and that the "true value" for the HEU case is b ≈ ¼ to ½. Normalizing to the upper end of Serber's range for HEU, we choose as our best estimate of b in equation (6.60), for

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[9] For Y expressed in joules: $(10^8 \text{ shakes/s})^2(10^3 \text{ g/kg})(10^{-7} \text{ j/erg}) = 10^{12}$; and for Y expressed in kt: $(10^{12} \text{ j})(0.238846 \text{ calories/j})(10^{-12} \text{ kt/calorie}) = 0.24$.

[10] $[(6.022\text{x}10^{23} \text{ nuclei/mol})(180 \text{ MeV/fission})(10^6 \text{ ev/MeV})(1.60219\text{x}10^{-19} \text{ j/ev})]/(10^{-7} \text{ j/erg})(235 \text{ g/mol})/ = 7.39\text{x}10^{17}$ ergs/gU-235; $= 7.454\text{x}10^{17}$ ergs/gU-233: and $= 7.267\text{x}10^{17}$ ergs/gPu-239.

[11] Serber, *The Los Alamos Primer*, (1992), p. 42.

HEU, U-233, and WGPu:

	b
HEU	0.5
U-233	0.8
WGPu	1.0

Table 1. Critical Mass Data for Estimating the Yield and Efficiency of Bare Nuclear Assemblies.[12]

	ρ (g/cm^3)	M_o (kg)	R_o (cm)	$(\rho R)_c$ (g/cm^2)	$\alpha_{\infty,o}$ (1/shake)	$(R_o\alpha_{\infty,o})^2$ (cm/shake)^2
αWGPu (5% Pu-240)	19.5	10.50	5.05	98.419	2.73	190
δWGPu (5% Pu-240)	15.660	16.29	6.285	98.419	2.19	190
WGU (95% U-235)	18.554	51.35	8.71	161.61	1.1	92
WGU (93.5% U-235)	18.74	52.42	8.74	165.8	1.1	92
WGU (93.86% U-235)	18.81	51.94	8.70	163.70	1.1	92
99.4% U-233	18.218	16.20	5.965	108.68	2.09	155
98.11% U-233	18.42	16.53	5.984	110.22	2.09	156
$^{239}PuO_2$	3.0	342.7	30.1	90.3		

αWGPu : α-phase weapon-grade plutonium.
δWGPu : δ-phase weapon-grade plutonium.
WGU : weapon-grade uranium
ρ : density
M_o : bare critical mass at normal density
R_o : bare critical radius at normal density
$\alpha_{\infty,o}$: neutron multiplication in an infinite medium at normal density

[12] J. Ligou, "Neutron Kinetics of Highly Compressed Fissionable Pellets," *Nuclear Science and Engineering*, 63, 31-40 (1977); Randall K. Cole, Jr. and James H. Renken, "Analysis of the Microfission Reactor Concept," *Nuclear Science and Engineering*, 58, 345-353 (1975); James H. Renken, "Alternative Materials for Microfission Target Pellets," *Nuclear Science and Engineering*, 59, 442-444 (1976) [check date and volume]; H.C. Paxton, "Los Alamos Critical-Mass Data," Los Alamos Scientific Laboratory, LA-3067-MS, Rev., pp. 4, 50 (November, 1975).

Table 2. Rossi-α values calculated Using the DTF-IV Neutron Transport Computer Code.[13]

Bare Pu-239

ρR (g/cm^2)	$R\alpha$ (cm/shake)
98.419	0.0
150	9.7
200	18.5
300	34.8
400	50.6
500	66.1
600	81.2
700	95.7
800	110.4
900	126.2
1000	138.5

Bare U-233

ρR (g/cm^2)	$R\alpha$ (cm/shake)
108.68	0.0
150	6.7
200	14.2
300	28.8
400	41.8
500	54.5

[13] James H. Renken, "Alternative Materials for Microfission Target Pellets," *Nuclear Science and Engineering*, 59, 442-444 (1976) [check date and volume]; data read from graphs.

Table 3. Neutron Multiplication in an Infinite Medium, k_∞^*, where the Flux Distribution Corresponds to the Actual Neutron Spectrum of a Close-to-Critical Assembly.[14]

	k_∞^*
^{239}Pu	2.927
^{239}PuN	2.788
^{239}PuC	2.904
$^{239}PuO_2$	2.871
^{235}U	2.320
^{235}UN	2.191
$^{235}UC_2$	2.269
$^{235}UO_2$	2.263
^{233}U	2.551
^{233}UN	2.438
$^{233}UC_2$	2.522
$^{233}UO_2$	2.511

[14] Anil Kumar, M. Srinivasan, K, Subba Rao, "Characterization of Neutron Leakage Probability, k_{eff}, and Critical Core Surface Mass Density of Small Reactor Assemblies Through the Trombey Criticality Formula," *Nuc. Sci. Eng.*, 82, 155-164 (1982); these are DTF-IV code results using Hansen and Roach cross sections.

Table 4. Mean Lifetime of a Neutron in Delayed Critical Masses (Finite Media) at Normal Density, τ_o ,[15] and in a Infinite Media at Normal Density, $\tau_{a,o}$.[16]

Critical Assembly		τ_o (shakes)	$\tau_{a,o}$ (shakes)
Bare WGU	(93.5% U-235) (Godiva)	0.66	1.5
	(53% U-235)	1.20	
	(38% U-235)	1.73	
	(29% U-235)	2.16	
WGU (93.5% U-235) in a thick natural U reflector (Topsy)		2.0	
WGPu (5% Pu-240) (Jezebel)		0.35	1.0

[15] John D. Orndoff, "Prompt Neutron Periods of Metal Critical Assemblies," *Nuclear Science and Engineering*, 2, 450-460 (1957).

[16] Calculated from $\tau_{a.o} = k_\infty \tau_o \approx k_\infty^* \tau_o$, and using k_∞^* data from Tables 3.2, and τ_o from this table.